

Teaching Traffic Lights to Take Turns: Multi-Agent Reinforcement Learning for Mixed Motorcycle Traffic

Dimas Arya Saputra^{1*}, Wahyu Hidayat¹, Maya Kusumawardani²

¹ MGM Laboratory, Department of Informatics

² Faculty of Computer Science, a university in East Java, Indonesia

* Corresponding author

ABSTRACT

Traffic signal controllers are usually tuned for cars in neat lanes, while many Indonesian intersections are dominated by motorcycles that fill every gap. We extend a microscopic simulator with sub-lane motorcycle behavior calibrated from drone video and train cooperative signal agents on a 12 intersection corridor. Compared with the fixed-time plan in use, the learned controllers cut average delay by 19 percent and queue spillback events by 41 percent in simulation.

Keywords: reinforcement learning, traffic signal control, multi-agent systems, simulation

1. Introduction

Anyone who has waited at a red light in a sea of motorcycles knows the timing plan was not written for this. Motorcycles queue differently, start faster and squeeze into space that a car model treats as empty.

2. Method

We calibrate sub-lane movement from 6 hours of drone video at three intersections, then train one agent per intersection with a shared critic that sees its neighbors. Rewards combine delay and a penalty for queues that spill into the upstream intersection [1, 4].

To avoid learning tricks that only work in simulation, we randomize demand, driver aggressiveness and sensor dropouts during training.

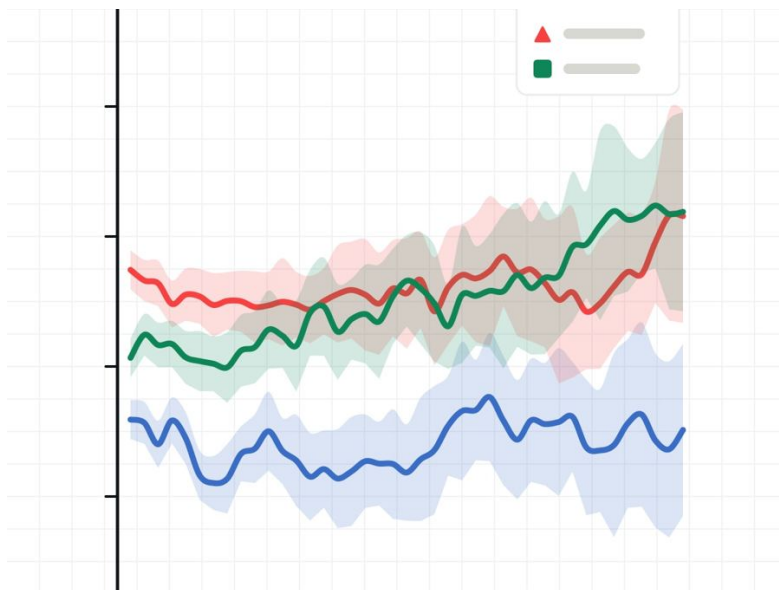


Figure 1. Average delay per episode during training with the spread across five seeds shaded.

3. Results

Cooperative agents beat both the fixed plan and independent agents, especially at peak hours (Table 1). The learned policies mostly shorten cycles when motorcycles dominate the queue.

Controller	Avg delay	Spillbacks per hour	Throughput
Fixed-time plan	74 s	9.6	baseline
Independent agents	66 s	7.1	+4%
Cooperative agents (ours)	60 s	5.7	+7%

Table 1. Corridor performance at the evening peak in simulation.

4. Discussion

Simulation gains rarely transfer one to one. We are talking with the city transport office about a shadow mode trial where the agent suggests timings but a person decides.

5. Conclusion

Modeling motorcycles properly changes what a good signal plan looks like, and learning helps find it.

References

[1] Wei, H. et al. IntelliLight: a reinforcement learning approach for intelligent traffic light control. KDD, 2018.

[2] Lopez, P. A. et al. Microscopic traffic simulation using SUMO. ITSC, 2018.

[3] Saputra, D. A. Calibrating sub-lane motorcycle behavior from drone video. Zemi working notes, 2025.

[4] Lowe, R. et al. Multi-agent actor-critic for mixed cooperative-competitive environments. NeurIPS, 2017.

[5] Chu, T. et al. Multi-agent deep reinforcement learning for large-scale traffic signal control. IEEE T-ITS, 2019.